

FATIH SULTAN MEHMET VAKIF UNIVERSITESI

Software Verification, Validation & Security Testing

TEST CASE DESIGN DOCUMENT

Smart Hospital Appointment System

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1. Introduction

This document defines all 88 test cases for the Smart Hospital Appointment System. Each test case includes the system requirement being verified, exact test data, step-by-step execution instructions, and expected results.

Column Definitions:

- Requirement: The system requirement this test case verifies — what the system MUST do.
- Test Data: The exact inputs, credentials, or data used during execution.
- Expected Result: The observable outcome confirming the requirement is satisfied.

Note on Automated Testing: Selenium WebDriver is used as a tool to automate execution of all 58 functional test cases programmatically via Chrome browser control. JDBC is used to inject required test data directly into the MySQL database before execution and clean up after each test. The automated scripts are not separate test cases — they execute the 58 functional test cases defined in this document.

2. Functional Test Cases

2.1 Authentication & Session

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-FUNC-001	Patient Login — Valid Credentials	Functional	System MUST authenticate patients with valid credentials and redirect to Patient Dashboard	URL: https://localhost:8080/login Username: patient1 Password: Patient@1234	1. Open login page 2. Enter username: patient1 3. Enter password: Patient@1234 4. Click LOGIN	Redirected to localhost:3000. 'Welcome back, patient!' displayed.
TC-FUNC-002	Admin Login — Valid Credentials	Functional	System MUST authenticate admins and redirect to Admin Dashboard with full management access	URL: https://localhost:8080/login Username: testadmin Password: Admin@1234	1. Open login page 2. Enter username: testadmin 3. Enter password: Admin@1234 4. Click LOGIN	Admin Dashboard displayed with Manage Users, Doctors, Departments tabs.
TC-FUNC-003	Doctor Login — Valid Credentials	Functional	System MUST authenticate doctors and redirect to Doctor Dashboard	URL: https://localhost:8080/login Username: doctor2 Password: Doctor@1234	1. Open login page 2. Enter username: doctor2 3. Enter password: Doctor@1234 4. Click LOGIN	Doctor Dashboard displayed with Patient, Schedule, Status, Actions columns.
TC-FUNC-004	Login — Invalid Credentials	Functional	System MUST reject invalid credentials and display error without revealing account details	Username: wronguser Password: wrongpass (both invalid)	1. Open login page 2. Enter username: wronguser 3. Enter password: wrongpass 4. Click LOGIN	'Invalid username or password.' displayed. User remains on login page.
TC-FUNC-005	Login — Empty Fields	Functional	System MUST prevent form submission when required fields are empty	Username: (empty) Password: (empty)	1. Open login page 2. Leave all fields empty 3. Click LOGIN	'Please fill this field' browser validation shown. Form not submitted.
TC-FUNC-017	Logout Functionality	Functional	System MUST terminate session on logout — back button must NOT restore dashboard	Logged in as: patient1 / Patient@1234	1. Login as patient1 2. Click Logout in navbar 3. Press browser Back button 4. Observe result	Redirected to landing page. Dashboard inaccessible. Session terminated.
TC-FUNC-019	Session Persistence After Refresh	Functional	System MUST maintain authenticated session after browser page refresh	Logged in as: patient1 / Patient@1234	1. Login as patient1 2. Navigate to dashboard 3. Press F5 to refresh	User remains logged in. No re-login required.
TC-FUNC-020	Login — Empty Form Direct Click	Functional	System MUST prevent login submission when all fields are empty	Username: (empty) Password: (empty)	1. Open login page 2. Click LOGIN directly without entering anything	'Please fill this field' shown. Form not submitted.
TC-FUNC-058	Password Case Sensitivity	Functional	System MUST treat passwords as case-sensitive — wrong case must be rejected	Username: patient1 Password: PATIENT@1234 (uppercase — wrong case)	1. Open login page 2. Enter username: patient1 3. Enter password: PATIENT@1234 4. Click LOGIN	'Invalid username or password.' shown. Login rejected.

2.2 Registration

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-FUNC-006	Registration — Valid Data	Functional	System MUST allow patient registration with valid data and redirect to login page	Full Name: Test Patient Username: newpatient1 Password: NewPass@1234 (unique username)	1. Open /register 2. Enter Full Name, Username, Password 3. Click Register	User registered. Redirected to login page. No errors.
TC-FUNC-007	Registration — Weak Password	Functional	System MUST reject passwords not meeting policy: 8+ chars, uppercase, number, special char	Full Name: Test User Username: testuser99 Password: 12345 (weak)	1. Open /register 2. Enter Name and Username 3. Enter Password: 12345 4. Click Register	'Password is too weak. Needs 8+ chars, 1 Upper, 1 Number, 1 Special' shown.
TC-FUNC-008	Registration — Duplicate Username	Functional	System MUST reject registration if chosen username already exists in the database	Full Name: Another User Username: patient1 (already exists) Password: Valid@1234	1. Open /register 2. Enter Full Name 3. Enter Username: patient1 4. Enter Password 5. Click Register	'Username is already taken!' error displayed.
TC-FUNC-029	Registration — Empty Full Name	Functional	System MUST reject registration when Full Name field is empty	Full Name: (empty) Username: newuser99 Password: Valid@1234	1. Open /register 2. Leave Full Name empty 3. Enter valid Username and Password 4. Click Register	'Please fill this field' shown. Registration prevented.
TC-FUNC-030	Registration — Empty Username	Functional	System MUST reject registration when Username field is empty	Full Name: Test User Username: (empty) Password: Valid@1234	1. Open /register 2. Enter Full Name 3. Leave Username empty 4. Enter Password 5. Click Register	'Please fill this field' shown. Registration prevented.
TC-FUNC-031	Registration — Empty Password	Functional	System MUST reject registration when Password field is empty	Full Name: Test User Username: newuser99 Password: (empty)	1. Open /register 2. Enter Full Name and Username 3. Leave Password empty 4. Click Register	'Please fill this field' shown. Registration prevented.
TC-FUNC-051	Registration — Invalid Characters in Name	Functional	System MUST reject registration with invalid special characters in Full Name	Full Name: test@#invalid! Username: testuser99 Password: Valid@1234	1. Open /register 2. Enter Full Name: test@#invalid! 3. Enter valid Username and Password 4. Click Register	Validation error displayed. Invalid characters rejected. No account created.
TC-FUNC-052	Registration — Username Too Short	Functional	System MUST enforce minimum username length of 4 characters	Full Name: Test User Username: ab (2 chars — too short) Password: Valid@1234	1. Open /register 2. Enter Full Name 3. Enter Username: ab 4. Enter Password 5. Click Register	Error shown: min 4 chars. Registration rejected.
TC-FUNC-053	Registration — Special Characters in Username	Functional	System MUST reject usernames containing special characters	Full Name: Test User Username: test@user! Password: Valid@1234	1. Open /register 2. Enter Full Name 3. Enter Username: test@user! 4. Enter Password 5. Click Register	Error: 'Letters and numbers only'. Registration rejected.
TC-FUNC-048	Password — Exact 8 Chars (Boundary)	Functional	System MUST accept passwords that exactly meet the minimum length of 8 characters	Full Name: Boundary User Username: boundaryuser1 Password: Valid@12 (exactly 8 chars)	1. Open /register 2. Enter Full Name and Username 3. Enter Password: Valid@12 4. Click Register	Registration succeeds. Password with exactly 8 characters accepted.
TC-FUNC-049	Username — Exact 4 Chars (Boundary)	Functional	System MUST accept usernames that exactly meet the minimum length of 4 characters	Full Name: Boundary User Username: abcd (exactly 4 chars) Password: Valid@1234	1. Open /register 2. Enter Full Name 3. Enter Username: abcd 4. Enter Password 5. Click Register	Registration succeeds. Username with exactly 4 characters accepted.

2.3 Patient Module

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-FUNC-009	Patient Views Departments	Functional	System MUST display available departments on Patient Dashboard	Logged in as: patient1 / Patient@1234 Dept in DB: Software Eng	1. Login as patient1 2. Navigate to Patient Dashboard 3. Observe departments list	Software Eng department displayed with available doctors.
TC-FUNC-010	Patient Books Appointment	Functional	System MUST allow patients to book appointments and store with PENDING status	Logged in as: patient1 / Patient@1234 Doctor: Dr. Salem Date: any future date Time: any slot	1. Login as patient1 2. Click Software Eng dept 3. Click Book Appointment for Dr. Salem 4. Select date and time 5. Click Confirm Booking	Booking confirmed. Appointment in Medical History with PENDING status.
TC-FUNC-011	Patient Views Medical History	Functional	System MUST display complete appointment history with status and prescription details	Logged in as: patient1 / Patient@1234 Appointment exists in DB	1. Login as patient1 2. Click History in navbar 3. Observe appointments list	Appointments shown with doctor, department, date, status, and prescription.
TC-FUNC-016	RBAC — Patient Cannot Access Admin	Functional	System MUST deny patients access to admin pages via URL manipulation	Logged in as: patient1 / Patient@1234 Target: http://localhost:3000/admin-dashboard	1. Login as patient1 2. Type in browser: http://localhost:3000/admin-dashboard 3. Press Enter	Access denied. Redirected to Patient Dashboard.
TC-FUNC-024	Patient Views Appointment Details	Functional	System MUST display full appointment details including prescriptions in Medical History	Logged in as: patient1 / Patient@1234 Existing appointment with prescription exists	1. Login as patient1 2. Click History 3. View appointment card	Full details: doctor, dept, date, status, Diagnosis, Medicine, Dosage.
TC-FUNC-028	Duplicate Appointment Prevention	Functional	System MUST prevent patients from booking same doctor at same date and time twice	Logged in as: patient1 / Patient@1234 Existing appointment at same time	1. Login as patient1 2. Try to book Dr. Salem at same date/time as existing 3. Click Confirm	'Failed to book. Please try again.' or error. No duplicate created.
TC-FUNC-054	Patient Views Empty Medical History	Functional	System MUST display appropriate message when patient has no appointments	Logged in as: newpatient1 (account with no appointments)	1. Login as newpatient1 2. Click History in navbar 3. Observe result	Empty state shown. No errors or crashes.
TC-FUNC-057	Patient Cannot Access Doctor Dashboard	Functional	System MUST deny patients access to doctor pages via URL manipulation	Logged in as: patient1 / Patient@1234 Target: http://localhost:3000/doctor-dashboard	1. Login as patient1 2. Type in browser: http://localhost:3000/doctor-dashboard 3. Press Enter	Access denied. Redirected to Patient Dashboard.
TC-FUNC-045	Book Appointment — Empty Date	Functional	System MUST reject booking when date field is empty	Logged in as: patient1 / Patient@1234 Date: (empty) Time: 10:00 AM	1. Login as patient1 2. Open Book Appointment 3. Leave date empty 4. Enter time 5. Click Confirm	'Please fill this field' shown. Booking prevented.
TC-FUNC-046	Book Appointment — Empty Time	Functional	System MUST reject booking when time field is empty	Logged in as: patient1 / Patient@1234 Date: any future date Time: (empty)	1. Login as patient1 2. Open Book Appointment 3. Enter date 4. Leave time empty 5. Click Confirm	'Please fill this field' shown. Booking prevented.
TC-FUNC-047	Book Appointment — Past Date	Functional	System MUST reject booking with a date in the past	Logged in as: patient1 / Patient@1234 Date: yesterday's date Time: 10:00 AM	1. Login as patient1 2. Open Book Appointment 3. Select yesterday's date 4. Click Confirm	Calendar disables past dates. Past date selection not permitted.
TC-FUNC-050	Book Appointment — Same Timeslot	Functional	System MUST reject booking if patient already has appointment at the same time	Logged in as: patient1 / Patient@1234 Existing appointment at same date/time	1. Login as patient1 2. Try to book different doctor at exact same date/time 3. Click Confirm	System rejects overlap. Error shown. No duplicate created.

2.4 Doctor Module

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-FUNC-015	Doctor Views Appointments	Functional	System MUST display all appointments assigned to the logged-in doctor	Logged in as: doctor2 / Doctor@1234 Appointment in DB: doctor_id=2	1. Login as doctor2 2. Navigate to Doctor Dashboard 3. Observe table	Table shown with Patient, Schedule, Status, Actions columns.
TC-FUNC-021	Doctor Accepts Appointment	Functional	System MUST allow doctors to accept pending appointments — status changes to APPROVED	Logged in as: doctor2 / Doctor@1234 PENDING appointment in DB	1. Login as doctor2 2. Find PENDING appointment 3. Click Accept 4. Observe status change	Status changes to APPROVED. Visible in patient Medical History.
TC-FUNC-022	Doctor Rejects Appointment	Functional	System MUST allow doctors to reject pending appointments — status changes to REJECTED	Logged in as: doctor2 / Doctor@1234 PENDING appointment in DB	1. Login as doctor2 2. Find PENDING appointment 3. Click Reject 4. Observe status change	Status changes to REJECTED. Record remains in system.
TC-FUNC-023	Doctor Writes Prescription	Functional	System MUST allow doctors to write and save prescriptions using OpenFDA search	Logged in as: doctor2 / Doctor@1234 APPROVED appointment exists Diagnosis: test Medicine: ttt Dosage: 500	1. Login as doctor2 2. Click Prescribe on APPROVED 3. Enter Diagnosis, Medicine, Dosage 4. Click Save Prescription	'Prescription Saved!' shown. Visible in patient Medical History.
TC-FUNC-027	Doctor Dashboard Refresh	Functional	System MUST reload latest appointments when Refresh button is clicked	Logged in as: doctor2 / Doctor@1234	1. Login as doctor2 2. Navigate to Doctor Dashboard 3. Click Refresh button (top right)	Appointments list reloaded. New appointments now visible.
TC-FUNC-042	Prescription — Empty Diagnosis	Functional	System MUST reject prescription when Diagnosis field is empty	Logged in as: doctor2 / Doctor@1234 Diagnosis: (empty) Medicine: aspirin Dosage: 500mg	1. Login as doctor2 2. Click Prescribe 3. Leave Diagnosis empty 4. Enter Medicine and Dosage 5. Click Save	'Please fill this field' shown. Prescription not saved.
TC-FUNC-043	Prescription — Empty Medicine	Functional	System MUST reject prescription when Medicine field is empty	Logged in as: doctor2 / Doctor@1234 Diagnosis: test Medicine: (empty) Dosage: 500mg	1. Login as doctor2 2. Click Prescribe 3. Enter Diagnosis 4. Leave Medicine empty 5. Enter Dosage 6. Click Save	'Please fill this field' shown. Prescription not saved.
TC-FUNC-044	Prescription — Empty Dosage	Functional	System MUST reject prescription when Dosage field is empty	Logged in as: doctor2 / Doctor@1234 Diagnosis: test Medicine: aspirin Dosage: (empty)	1. Login as doctor2 2. Click Prescribe 3. Enter Diagnosis and Medicine 4. Leave Dosage empty 5. Click Save	'Please fill this field' shown. Prescription not saved.
TC-FUNC-055	Doctor Cannot Access Admin Panel	Functional	System MUST deny doctors access to admin-only pages	Logged in as: doctor2 / Doctor@1234 Target: http://localhost:3000/admin-dashboard	1. Login as doctor2 2. Type in browser: http://localhost:3000/admin-dashboard 3. Press Enter	Access denied. Redirected to Doctor Dashboard.

2.5 Admin Module

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-FUNC-012	Admin Creates Doctor	Functional	System MUST allow admins to create new doctor accounts visible in Doctors List	Logged in as: testadmin / Admin@1234 Name: Dr. Test Username: drtest1 Password: Doctor@1234! Specialization: Surgery Dept: Software Eng	1. Login as testadmin 2. Go to Manage Doctors 3. Fill Create Doctor form 4. Click Create	New doctor appears in Doctors List immediately.
TC-FUNC-013	Admin Creates Patient	Functional	System MUST allow admins to create patient accounts from Admin Dashboard	Logged in as: testadmin / Admin@1234 Name: New Patient Username: newpt99 Password: Patient@1234	1. Login as testadmin 2. Go to Manage Users 3. Fill Create Patient form 4. Click Create	New patient appears in Users List with PATIENT role.
TC-FUNC-014	Admin Deletes Patient	Functional	System MUST allow admins to permanently remove patient accounts	Logged in as: testadmin / Admin@1234 Target: newpatient1	1. Login as testadmin 2. Go to Manage Users 3. Click Delete next to newpatient1 4. Confirm	Patient removed from Users List.
TC-FUNC-018	Admin Manages Departments	Functional	System MUST provide admins with department viewing and management capabilities	Logged in as: testadmin / Admin@1234	1. Login as testadmin 2. Click Manage Departments tab 3. Observe list	Departments displayed with management options.
TC-FUNC-025	Admin Edits Doctor	Functional	System MUST allow admins to update doctor information with immediate effect	Logged in as: testadmin / Admin@1234 Target: any doctor New Specialization: Updated Surgery	1. Login as testadmin 2. Manage Doctors 3. Click Edit 4. Update Specialization 5. Save	Changes reflected immediately in Doctors List.
TC-FUNC-026	Admin Edits Patient	Functional	System MUST allow admins to update patient information with immediate effect	Logged in as: testadmin / Admin@1234 Target: any patient New Name: Updated Name	1. Login as testadmin 2. Manage Users 3. Click Edit 4. Update Full Name 5. Save	Changes reflected immediately in Users List.
TC-FUNC-056	Admin Cannot Access Doctor Dashboard	Functional	System MUST deny admins access to doctor-only pages	Logged in as: testadmin / Admin@1234 Target: http://localhost:3000/doctor-dashboard	1. Login as testadmin 2. Type in browser: http://localhost:3000/doctor-dashboard 3. Press Enter	Access denied. Redirected to Admin Dashboard.
TC-FUNC-032	Create Doctor — Empty Name	Functional	System MUST reject doctor creation when Full Name is empty	Logged in as: testadmin / Admin@1234 Name: (empty) Username: drtest2 Password: Doctor@1234	1. Login as testadmin 2. Manage Doctors 3. Leave Name empty 4. Fill others 5. Click Create	'Please fill this field' shown. Doctor not created.
TC-FUNC-033	Create Doctor — Empty Username	Functional	System MUST reject doctor creation when Username is empty	Logged in as: testadmin / Admin@1234 Name: Dr. Test Username: (empty) Password: Doctor@1234	1. Login as testadmin 2. Manage Doctors 3. Leave Username empty 4. Fill others 5. Click Create	'Please fill this field' shown. Doctor not created.
TC-FUNC-034	Create Doctor — Empty Password	Functional	System MUST reject doctor creation when Password is empty	Logged in as: testadmin / Admin@1234 Name: Dr. Test Username: drtest3 Password: (empty)	1. Login as testadmin 2. Manage Doctors 3. Leave Password empty 4. Fill others 5. Click Create	'Please fill this field' shown. Doctor not created.
TC-FUNC-035	Create Doctor — Empty Specialization	Functional	System MUST reject doctor creation when Specialization is empty	Logged in as: testadmin / Admin@1234 Specialization: (empty)	1. Login as testadmin 2. Manage Doctors 3. Leave Specialization empty 4. Fill others 5. Click Create	'Please fill this field' shown. Doctor not created.
TC-FUNC-036	Create Doctor — Empty Department	Functional	System MUST reject doctor creation when Department is not selected	Logged in as: testadmin / Admin@1234 Department: (not selected)	1. Login as testadmin 2. Manage Doctors 3. Leave Department unselected 4. Fill others 5. Click Create	'Please fill this field' shown. Doctor not created.
TC-FUNC-037	Create Patient — Empty Name	Functional	System MUST reject patient creation when Full Name is empty	Logged in as: testadmin / Admin@1234 Name: (empty) Username: newpt100 Password: Patient@1234	1. Login as testadmin 2. Manage Users 3. Leave Name empty 4. Fill others 5. Click Create	'Please fill this field' shown. Patient not created.
TC-FUNC-038	Create Patient — Empty Username	Functional	System MUST reject patient creation when Username is empty	Logged in as: testadmin / Admin@1234 Name: New Patient Username: (empty) Password: Patient@1234	1. Login as testadmin 2. Manage Users 3. Leave Username empty 4. Fill others 5. Click Create	'Please fill this field' shown. Patient not created.
TC-FUNC-039	Create Patient — Empty Password	Functional	System MUST reject patient creation when Password is empty	Logged in as: testadmin / Admin@1234 Name: New Patient Username: newpt101 Password: (empty)	1. Login as testadmin 2. Manage Users 3. Leave Password empty 4. Fill others 5. Click Create	'Please fill this field' shown. Patient not created.
TC-FUNC-040	Edit Doctor — Clear Specialization	Functional	System MUST reject saving doctor profile when Specialization is cleared	Logged in as: testadmin / Admin@1234 Target: any doctor	1. Login as testadmin 2. Manage Doctors 3. Click Edit 4. Clear Specialization 5. Click Save	'Please fill this field' shown. Specialization cannot be empty.
TC-FUNC-041	Edit Patient — Clear Name	Functional	System MUST reject saving patient profile when Full Name is cleared	Logged in as: testadmin / Admin@1234 Target: any patient	1. Login as testadmin 2. Manage Users 3. Click Edit 4. Clear Full Name 5. Click Save	'Please fill this field' shown. Full Name cannot be empty.

3. Performance Test Cases

All performance tests executed using Apache JMeter 5.6.3. Target: <https://localhost:8080/login>

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-PERF-001	Response Time — 10 Concurrent Users	Performance	System MUST respond within 2 seconds under normal load of 10 concurrent users	Tool: JMeter 5.6.3 Threads: 10 Ramp-up: 10s URL: GET https://localhost:8080/login	1. Open JMeter 2. Create Thread Group: 10 users, ramp-up=10s 3. Add HTTP GET Request 4. Add Summary Report 5. Run	Avg < 2s. Error 0%. All 10 requests succeed.
TC-PERF-002	Load Test — 50 Concurrent Users	Performance	System MUST remain stable with response under 5 seconds under 50 concurrent users	Tool: JMeter 5.6.3 Threads: 50 Ramp-up: 30s	1. Open JMeter 2. Thread Group: 50 users, ramp-up=30s 3. HTTP Request and Summary Report 4. Run	Avg < 5s. Error < 5%. System stable.
TC-PERF-003	Stress Test — 100 Concurrent Users	Performance	System MUST handle 100 concurrent users without server errors	Tool: JMeter 5.6.3 Threads: 100 Ramp-up: 60s	1. Open JMeter 2. Thread Group: 100 users, ramp-up=60s 3. HTTP Request and Summary Report 4. Run and observe	Error 0%. No crashes. Breaking point documented.
TC-PERF-004	Login Endpoint — 20 POST Users	Performance	System MUST handle 20 concurrent POST login requests within 3 seconds each	Tool: JMeter 5.6.3 Threads: 20 Method: POST URL: https://localhost:8080/login	1. Open JMeter 2. Thread Group: 20 users 3. HTTP POST to login 4. Run 60s 5. Summary Report	Response < 3s. Error 0%.
TC-PERF-005	Patient Dashboard Load — 10 Users	Performance	System MUST load Patient Dashboard within 3 seconds for 10 concurrent users	Tool: JMeter 5.6.3 Threads: 10 URL: http://localhost:3000	1. Open JMeter 2. Thread Group: 10 users 3. HTTP GET to localhost:3000 4. Summary Report	Dashboard loads < 3s. Error 0%.
TC-PERF-006	Booking Endpoint — 20 Users	Performance	System MUST handle 20 concurrent booking requests without errors	Tool: JMeter 5.6.3 Threads: 20 URL: https://localhost:8080/login	1. Open JMeter 2. Thread Group: 20 users 3. HTTP Request to booking endpoint 4. Run 60s	Response < 4s. Error 0%.
TC-PERF-007	Admin Dashboard Load — 15 Users	Performance	System MUST load Admin Dashboard within 3 seconds for 15 concurrent users	Tool: JMeter 5.6.3 Threads: 15 URL: https://localhost:8080/login	1. Open JMeter 2. Thread Group: 15 users 3. HTTP GET to admin endpoint 4. Run 30s	Admin dashboard < 3s. Error 0%.

4. Usability Test Cases

Method: Think-Aloud Protocol with 5 independent testers: Tester 1, Tester 2, Tester 3, Tester 4, Tester 5.

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-USA-001	Patient Finds Booking Without Help	Usability	At least 4/5 testers MUST locate booking flow within 3 minutes without assistance	Tester: unfamiliar with system Logged in as: patient1 Dept available: Software Eng	Give scenario: 'You are a new patient. Find a doctor and book an appointment.' Observe navigation. Record time and Think-Aloud comments.	4/5 testers locate Book Appointment within 3 minutes without help.
TC-USA-002	Admin Creates Doctor Without Help	Usability	At least 4/5 testers MUST complete doctor creation within 2 minutes without training	Tester: unfamiliar with admin panel Logged in as: testadmin Task: Add Dr. Khalid in Cardiology	Give scenario: 'Add a new doctor named Dr. Khalid in Cardiology.' Observe navigation and form completion. Record Think-Aloud.	4/5 testers complete task in < 2 minutes without assistance.
TC-USA-003	New User Finds Register Link	Usability	At least 4/5 testers MUST find Register link within 30 seconds without assistance	Tester: sees login page for first time URL: https://localhost:8080/login	Give scenario: 'You are new and want to create an account. Please register.' Present login page. Observe and record Think-Aloud.	4/5 testers find 'Register Here' within 30 seconds.
TC-USA-004	Doctor Dashboard Clarity	Usability	At least 4/5 testers MUST identify pending appointment and available actions within 1 minute	Tester: logged in as doctor2 One PENDING appointment in dashboard	Give scenario: 'You have a new appointment. Find it and check patient details.' Observe on Doctor Dashboard.	4/5 testers identify appointment within 1 minute. Columns self-explanatory.
TC-USA-005	Overall Satisfaction Rating	Usability	System MUST achieve average satisfaction >= 3.5/5 across navigation, clarity, overall	5 independent testers Rating scale: 1 to 5	After all tasks ask each tester: 1. Rate navigation (1-5) 2. Rate clarity (1-5) 3. Rate overall (1-5) 4. Open comments	Average rating >= 3.5/5 across all three dimensions.
TC-USA-006	Visual Design & Color Consistency	Usability	4/5 testers MUST rate visual consistency >= 3.5/5. No tester reports confusing colors	All pages: Login, Patient/Doctor/Admin Dashboard, History 5 testers observe	1. Tester navigates all pages 2. Observe color scheme consistency 3. Note confusing colors using Think-Aloud 4. Rate consistency 1-5	Average rating >= 3.5/5. No confusing color clashes identified.
TC-USA-007	Typography & Readability	Usability	4/5 testers MUST rate readability >= 3.5/5 and confirm text readable without zooming	All pages visible to each role 5 testers observe during navigation	1. Tester reads headings, labels, table content on all pages 2. Note any unreadable text 3. Rate readability 1-5 4. Record Think-Aloud	All text readable without zooming. Avg rating >= 3.5/5.
TC-USA-008	Layout & Navigation Consistency	Usability	All 5 testers MUST confirm navbar always accessible. Avg layout rating >= 3.5/5	All authenticated pages for all 3 roles 5 testers observe	1. Login as each role and navigate all pages 2. Confirm navbar and Logout visible on every page 3. Note any layout inconsistencies 4. Rate 1-5	Navbar present on all pages. Logout always accessible. Avg >= 3.5/5.
TC-USA-009	Error Message Clarity	Usability	4/5 testers MUST rate error message clarity >= 3/5 and confirm they understood action needed	Error scenarios: wrong password, duplicate username, weak password, failed booking	1. Trigger each error scenario 2. Read error message 3. Rate clarity 1-5 4. Think-Aloud: 'Do I know what to do next?'	Error messages explain the issue. 4/5 testers understand corrective action.
TC-USA-010	Responsive Feedback & Loading	Usability	4/5 testers MUST confirm each action produces visible feedback within 2 seconds	Actions: Login, Register, Save Prescription, Logout 5 testers observe	1. Perform each action 2. Observe immediate visual response 3. Note silent failures 4. Rate feedback quality 1-5	Each action produces immediate feedback: success message, redirect, or loading indicator.

5. Security Test Cases

TC ID	Objective	Type	Requirement	Test Data	Test Steps	Expected Result
TC-SEC-001	SQL Injection on Login	Security	System MUST use parameterized queries to prevent SQL injection on all input fields	Injection payload: Username: admin' OR '1'=1 Password: anything URL: https://localhost:8080/login	1. Open login page 2. Enter username: admin' OR '1'=1 3. Enter password: anything 4. Click LOGIN 5. Observe	'Invalid username or password.' returned. No DB error. User NOT authenticated.
TC-SEC-002	XSS Attack on Registration	Security	System MUST sanitize inputs — scripts must NOT execute in the browser	Full Name: <script>alert('XSS')</script> Username: xsstest1 Password: Test@1234	1. Open /register 2. Enter Full Name: <script>alert('XSS')</script> 3. Enter valid Username and Password 4. Click Register 5. Observe	Script does NOT execute. No alert popup. Form rejects or sanitizes input.
TC-SEC-003	OWASP ZAP Automated Scan	Security	System MUST have no Critical unmitigated vulnerabilities in OWASP ZAP automated scan	Tool: OWASP ZAP 2.17 Target: https://localhost:8080 Scan: Automated (Spider + Active)	1. Open OWASP ZAP 2. Set target to https://localhost:8080 3. Run Automated Scan 4. Wait for completion 5. Review findings	Scan completes. All findings documented. No Critical vulnerabilities unmitigated.
TC-SEC-004	Brute Force Protection	Security	System MUST lock account or add delay after 5 consecutive failed login attempts	Username: patient1 Password: wrongpass123 10 consecutive failed attempts	1. Open login page 2. Enter patient1 / wrongpass123 3. Click LOGIN 10 times 4. Observe system behavior	After 5 failures: logout OR delay OR CAPTCHA triggered.
TC-SEC-005	Cookie Security — HttpOnly Flag	Security	Session cookie MUST have HttpOnly=true and Secure=true to prevent XSS hijacking	Logged in as: patient1 Browser: Chrome Tool: DevTools (Fn+F12)	1. Login as patient1 2. Open DevTools 3. Application > Cookies 4. Find JSESSIONID 5. Check attributes	JSESSIONID has HttpOnly=true and Secure=true confirmed.
TC-SEC-006	Unauthorized Admin Access by Patient	Security	System MUST deny patients access to admin pages regardless of URL manipulation	Logged in as: patient1 Target: http://localhost:3000/admin-dashboard	1. Login as patient1 2. Type in browser: http://localhost:3000/admin-dashboard 3. Press Enter	Access denied. Redirected to Patient Dashboard or 403 error.
TC-SEC-007	Password Strength Enforcement	Security	System MUST reject all passwords not meeting policy: 8+ chars, uppercase, number, special	Test passwords: - 123456 (too short) - password (no uppercase/special) - abc (too short)	1. Open /register 2. Try Password: 123456 3. Try Password: password 4. Try Password: abc	All weak passwords rejected. Error: 'Password is too weak...' shown for each.
TC-SEC-008	Session Invalidation After Logout	Security	System MUST completely invalidate session on logout — back button must NOT restore access	Logged in as: patient1 / Patient@1234	1. Login as patient1 2. Confirm on dashboard 3. Click Logout 4. Press browser Back button 5. Observe	Session invalidated. Back button shows landing page not dashboard.
TC-SEC-009	CSRF Protection Check	Security	System MUST validate CSRF tokens on all state-changing POST requests	Source code: SecurityConfig.java Tool: IntelliJ IDEA inspection	1. Open SecurityConfig.java 2. Find securityFilterChain 3. Check .csrf() configuration 4. Determine enabled or disabled	CSRF protection MUST be active. .csrf(disable) must NOT exist.
TC-SEC-010	Directory Traversal Prevention	Security	System MUST reject traversal attempts and not expose server file system	Attack URLs: https://localhost:8080/./etc/passwd https://localhost:8080/./../windows/system32	1. Open browser 2. Navigate to each attack URL 3. Observe response	400 or 404 returned. No system files exposed. Spring Boot rejects by default.
TC-SEC-011	No Sensitive Data in URL	Security	System MUST NOT transmit passwords, tokens, or personal data in URL parameters	Logged in as: patient1 Pages: Dashboard, History, Departments	1. Login as patient1 2. Navigate all pages 3. Inspect URL bar on each page	All URLs contain only route paths. No passwords or tokens in any URL.
TC-SEC-012	No Cross-Patient Data via Admin	Security	Admin edit form MUST show only account info — no medical history or prescriptions	Logged in as: testadmin Two patients exist: patient1, newpatient1	1. Login as testadmin 2. Manage Users 3. Click Edit on patient1 4. Observe form fields	Only name and username visible. No medical history or prescriptions shown.
TC-SEC-013	Account Locked After Brute Force	Security	System MUST prevent login even with correct password during lockout period	Username: patient1 Wrong password: wrongpass 10 consecutive failures Then: correct password Patient@1234	1. Fail login 10 times with patient1 / wrongpass 2. After lockout triggered, enter correct password 3. Observe result	Login rejected during lockout. Account temporarily locked message shown.

6. Test Case Summary

Testing Type	Count	ID Range
Functional Testing	58	TC-FUNC-001 to TC-FUNC-058
Performance Testing	7	TC-PERF-001 to TC-PERF-007
Usability Testing	10	TC-USA-001 to TC-USA-010
Security Testing	13	TC-SEC-001 to TC-SEC-013
TOTAL	88	Exceeds 20 minimum requirement

Note: Selenium WebDriver is used as an automation tool to execute all 58 functional test cases programmatically. JDBC is used for test data setup and cleanup. These automated scripts do not represent additional test cases beyond the 88 defined above.